



Perry Nuclear Power Plant
Rod Penfield
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L-24-168
July 15, 2024

10 CFR 50.73(a)(2)(i)(A)

ATTN: Document Control Desk
U.S. Nuclear Regulatory Commission
Washington, DC 20555-0001

Subject: Perry Nuclear Power Plant, Unit 1
Docket Number 50-440, License Number NPF-58
Licensee Event Report 2024-003-00

Enclosed is Licensee Event Report (LER) 2024-003-00, "*Technical Specification Required Shutdown Due to Increase in RCS Unidentified Leakage*." This event is being reported pursuant to 10 CFR 50.73(a)(2)(i)(A).

There are no regulatory commitments contained in this letter or its enclosure. If there are any questions or if additional information is required, please contact Mr. Robert W. Oesterle, Manager, Regulatory Compliance, at (419) 321-7462.

Sincerely,

A handwritten signature in black ink, appearing to read "Rod L. Penfield".

Rod L. Penfield

Enclosure: LER 2024-003-00

cc: NRC Region III Administrator
NRC Resident Inspector
NRR Project Manager



LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form
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Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Services Branch (T-6 A10M), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. Facility Name

Perry Nuclear Power Plant, Unit 1

☒ 050
☐ 052

2. Docket Number

440

3. Page

1 OF 4

4. Title:

Technical Specification Required Shutdown Due to Increase in RCS Unidentified Leakage

5. Event Date

Month	Day	Year
05	23	2024

6. LER Number

Year	Sequential Number	Rev No.
2024	003	00

7. Report Date

Month	Day	Year
07	15	2024

8. Other Facilities Involved

Facility Name	☐ 050	Docket Number
Facility Name	☐ 052	Docket Number

9. Operating Mode

1

10. Power Level

40

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)

10 CFR Part 20	☐ 20.2203(a)(2)(vi)	10 CFR Part 50	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.1200(a)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	☐ 73.1200(b)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	☐ 73.1200(c)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)	☐ 73.1200(d)
☐ 20.2203(a)(2)(i)	10 CFR Part 21	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	10 CFR Part 73	☐ 73.1200(e)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.77(a)(1)	☐ 73.1200(f)
☐ 20.2203(a)(2)(iii)		☒ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(2)(i)	☐ 73.1200(g)
☐ 20.2203(a)(2)(iv)		☐ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(ii)	☐ 73.1200(h)
☐ 20.2203(a)(2)(v)		☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)		

☐ OTHER (Specify here, in abstract, or in NRC Form 366A).

12. Licensee Contact for this LER

Licensee Contact

Chad D. Atkinson, Regulatory Compliance

Phone Number (Include area code)

(419) 321-7844

13. Complete One Line for Each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS

14. Supplemental Report Expected

☒ No ☐ Yes (If yes, complete 15. Expected Submission Date)

15. Expected Submission Date

Month	Day	Year

16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)

While operating in MODE 1 at 100% power on May 23, 2024 at 0000, Perry Nuclear Power Plant (PNPP) operators entered Technical Specification (TS) 3.4.5 due to the unidentified reactor coolant system (RCS) leak rate exceeding 5 gpm. PNPP operators commenced a manual reactor shutdown in accordance with site procedures at 0053 on May 23, 2024. Unable to reduce the unidentified RCS leakage, the shutdown transitioned to a TS required shutdown at 0400 and reported as Event Number 57136. The required safety systems and shutdown equipment performed as expected, resulting in an uncomplicated shutdown. The source of the unidentified RCS leakage was the reactor recirculation loop 'A' suction valve packing leak-off line via a failed sight glass flange. The leak-off line was isolated and sight glass repaired.

The root causes were failing to maintain and operate the leak detection system leak-off lines as designed because of unrecognized potential consequences from a failure in inaccessible areas and not prioritizing permanent repairs for long standing equipment degradation during outage scoping.

This event is being reported pursuant to 10 CFR 50.73(a)(2)(i)(A) as completion of a plant shutdown required by the Technical Specifications.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

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1. FACILITY NAME Perry Nuclear Power Plant Unit 1	☒ 050	2. DOCKET NUMBER 440	3. LER NUMBER		
	☐ 052		YEAR 2024	SEQUENTIAL NUMBER - 003	REV NO. - 00

NARRATIVE

Energy Industry Identification System (EIIIS) codes are identified in the text as [XX].

Background:

The Perry Nuclear Power Plant (PNPP) reactor coolant recirculation system [AD] is designed to provide a forced coolant flow through the core to remove heat from the fuel. The pressure containing components of the reactor coolant system (RCS) and the portions of connecting systems out to and including the isolation valves define the reactor coolant pressure boundary (RCPB). During plant life, the joint and valve interfaces can produce varying amounts of reactor coolant leakage through either normal operational wear or mechanical deterioration. Limits on RCS operational leakage are required to ensure appropriate action is taken before the integrity of the RCPB is impaired. The safety significance of leaks from the RCPB varies widely depending on the source, rate, and duration; therefore, detection of leakage in the drywell is necessary. Methods for quickly separating the identified LEAKAGE from the unidentified LEAKAGE are necessary to provide the operators quantitative information to permit them to take corrective action should a leak occur detrimental to the safety of the facility or the public. A limited amount of leakage inside the drywell is expected from auxiliary systems that cannot be made 100% leak tight. Leakage from these systems should be detected and isolated from the drywell atmosphere, if possible, so as not to mask RCS operational LEAKAGE detection. Twenty-one drywell valves, including the reactor coolant recirculation pump suction valves [AD-V], have packing leak-off lines routed to the drywell equipment drain sump [WK]. Leakage directed to the drywell equipment drain sump is considered identified leakage while leakage collected through the floor drains to the floor drain sump is unidentified. Both leakage rates display and alarm in the Control Room. Each valve packing leak-off line has a temperature transmitter [AD-TT] that provides an alarm in the Control Room on high temperature in the line. A manually operated solenoid valve [AD-TSV] can then be closed from the Control Room to isolate the line. Located in the drywell and inaccessible while at power, the leak-off lines have a manual isolation valve [AD-ISV] upstream and a sight glass [AD-FG] downstream of the solenoid valves.

Technical Specifications:

PNPP Technical Specification (TS) Limiting Condition for Operation (LCO) 3.4.5, "RCS Operational LEAKAGE" applies in MODES 1, 2, and 3 and requires: no pressure boundary LEAKAGE; less than or equal to 5 gpm unidentified LEAKAGE; less than or equal to 30 gpm total LEAKAGE over the previous 24 hour period; and less than or equal to 2 gpm increase in unidentified LEAKAGE within the previous 24 hour period in MODE 1. When unidentified or total leakage is not within the limits, TS 3.4.5 Action B.1 requires the leakage to be within limits in 4 hours, and Action C.1 requires verification that the unidentified LEAKAGE increase is not service sensitive austenitic material within 4 hours. If the required actions and associated completion times are not met, TS 3.4.5 Action D.1 requires the plant to be placed in MODE 3 within 12 hours, and Action D.2 requires the plant to be placed in MODE 4 within 36 hours.

DESCRIPTION OF EVENT:

With PNPP operating in MODE 1 and 100 percent power, the drywell unidentified rate of change high alarm was received at 0000 on May 23, 2024. The drywell floor drain sump rate indicated 7.7 gpm. Condition B of TS 3.4.5 was entered to reduce the unidentified leakage to less than or equal to 5 gpm within 4 hours and lowering reactor power to achieve MODE 3 was started, in accordance with site procedures. Subsequently, the operations crew received the valve stem leak-off temperature alarm, indicating at least one of the 21 drywell valve leak-off line temperatures were greater than or equal to 180 degrees F. Four lines were identified as potential sources. Simple troubleshooting found none of the four manually operated solenoid valves functioned. At 0400 on

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NARRATIVE

May 23, 2024, the drywell floor drain sump rate of 5.6 gpm was still above the TS operational limit; therefore, TS 3.4.5 Condition D was entered with required actions to place the plant in MODE 3 in 12 hours and MODE 4 in 36 hours. On May 23, 2024, PNPP entered MODE 3 at 1338 hours and MODE 4 at 2134 hours. During the initial drywell entry at 1124 hours on May 23, 2024, no RCPB leakage was identified; however, the reactor recirculation loop 'A' suction valve packing leak-off line sight glass flange was found failed. The failed flange allowed the reactor recirculation loop 'A' suction valve packing leakage to enter the drywell floor drain sump instead of the drywell equipment drain sump and increase the unidentified leakage.

CAUSE OF EVENT:

Reactor recirculation loop 'A' suction valve packing degradation was initially identified because of high stem leak-off line temperature during the 2013 refueling outage start-up. The direct cause of the unidentified leakage increase was failure to close the reactor recirculation loop 'A' suction valve packing leak-off line solenoid valve in 2013 through 2024 resulting in challenging downstream piping performance, causing sight glass flange leakage, and immediately exceeding unidentified RCS leakage TS limits.

The root causes of this event were:

- 1) Failing to maintain and operate the leak detection system leak-off lines as designed because of unrecognized potential consequences from a failure in inaccessible areas.
- 2) Inadequate risk recognition and consequence-bias during the Outage Scope Review Board and Engineering Review process led to not prioritizing permanent repairs for long standing equipment degradation.

ANALYSIS OF EVENT:

The operating crew responded correctly to the event. The applicable abnormal response procedure was properly entered, and performance met expectations. There were no safety systems inoperable, and no safety system failures related to this event. Since no RCPB leakage was identified, there were no actual safety consequences from this event. The leakage from the reactor recirculation loop 'A' suction valve packing leak-off line sight glass flange was contained within the drywell, and the plant reached a safe shutdown condition. There were no system failures that prevented the safe shutdown of the plant. It is therefore concluded that even if a design basis accident had occurred concurrent with this event, all safety systems would have operated to safely mitigate the event. Based on the above considerations, the safety significance of this event is very low, and the event did not pose a threat to the health and safety of the public or plant personnel.

Reportability Discussion:

The initiation of any nuclear plant shutdown required by the plant's Technical Specifications is reportable within four hours of the event in accordance with 10 CFR 50.72(b)(2)(i). Since a manual shutdown transitioned to a TS 3.4.5 required shutdown on May 23, 2024, at 0400 hours, the event was reported to the NRC Operations Center at 0425 hours as Event Number 57136.

This event is being reported in accordance with 10 CFR 50.73(a)(2)(i)(A) as a plant shutdown required by the Technical Specifications.



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NARRATIVE

CORRECTIVE ACTIONS:

Completed Actions:

The increased unidentified RCS leakage was stopped by closing the recirculation pump 'A' suction valve packing leak-off line manual isolation valve on May 23, 2024 at approximately 1600 hours. Additionally, the manual solenoid valve was closed on May 24, 2024 at approximately 0200 hours, the sight glass was repaired on May 24, 2024 at approximately 0500 hours, and the recirculation pump 'A' suction valve was back seated.

Scheduled Actions:

Future corrective actions include: developing and implementing a site valve packing preventive maintenance program for inaccessible locations while online, modifying the leak detection system design and updating the drywell valve packing designs, and revising outage scoping documents with specific guidance for leaks in inaccessible areas while online.

PREVIOUS SIMILAR EVENTS:

There have been no PNPP Licensee Event Reports related to a plant shutdown required by Technical Specifications in the past three years.